

Log-Based Recovery & Checkpointing — DBMS

Definition: Log-Based Recovery maintains a chronological record of all database modifications on non-volatile storage (the **Log**) to restore database Atomicity and Durability following system crashes.

1. Detailed Technical Explanation

1. Write-Ahead Logging (WAL) Protocol

Before any data modification is written to disk, the corresponding log record **MUST** be flushed to non-volatile log storage.

Log Record Formats:

- **<T_n, start>** : Transaction **T_n** started.
- **<T_n, X, V₁, V₂>** : Transaction **T_n** changed item **X** from Old Value **V₁** to New Value **V₂**.
- **<T_n, commit>** : Transaction **T_n** committed.
- **<T_n, abort>** : Transaction **T_n** aborted.

2. Recovery Techniques

1. Deferred Update Technique (NO-UNDO / REDO Algorithm)

- Database disk writes are postponed until the transaction reaches its **COMMIT** point.
- **Recovery Procedure:**
- **REDO(T_n)** : Replays new values **V₂** for transactions with both **<T_n, start>** and **<T_n, commit>** in log.
- No **UNDO** required because uncommitted data was never written to main database blocks!

2. Immediate Update Technique (UNDO / REDO Algorithm)

- Database disk writes can occur while the transaction is still active.
- **Recovery Procedure:**
- **UNDO(T_n)** : Restores old values **V₁** for active transactions with **<T_n, start>** but NO **<T_n, commit>**.
- **REDO(T_n)** : Replays new values **V₂** for committed transactions.

3. Checkpointing

Checkpointing reduces recovery overhead by periodically flushing all dirty buffer blocks and active transaction logs to disk.

Checkpoint Record Outputted

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<T1, start> ... <T2, start> <CHECKPOINT {T1, T2}> ... <T2, commit> [CRASH!]

- During crash recovery, the system only needs to scan the log backwards up to the **most recent checkpoint**.

4. Core Concepts & Memory Keywords

- **WAL Protocol:** Write log to disk BEFORE writing database blocks.
- **REDO:** Replay operations (**V_2**) for committed transactions.
- **UNDO:** Reverse operations (**V_1**) for uncommitted / aborted transactions.
- **Checkpoint:** Periodic point flushing dirty RAM buffers to persistent disk.

5. Must-Write Points for Exams

- WAL guarantees Durability by ensuring committed transactions can be reconstructed via REDO log records.
- Deferred update requires only REDO operations during recovery; immediate update requires both UNDO and REDO.
- Checkpointing eliminates the need to scan the entire log file from the beginning of database creation.

6. Quick Recall Flow

Write Log Record to Disk (WAL) -> Modify Memory -> Checkpoint -> System Crash -> UNDO Active, REDO Committed